Parallel Programming: Assignment 1

Dense Matrix-Matrix Multiplication

There are many parallel algorithms for dense matrix-matrix multiplication. Explore and find a good parallel algorithm, implement it by yourself using MPI. Also, implement Cannon's algorithm dealt in the class. Perform experiments and compare the two algorithms for three matrix inputs and 3 number of processors.

3 Matrix inputs:

- Input 1: 6400×6400 matrices for A, B and C,
- Input 2: 12800×12800 matrices for A, B and C,
- Input 3: 25600×25600 matrices for A, B and C.

Number of processors: 16 and 64 processor cores.

The two matrices A and B should be read from file and the C matrix should be stored in file using MPI I/O interfaces. Also, compare with sequential matrix-matrix multiplication, plot the execution time and speedup curves, and prepare a report containing the method, graphs and observations.

Your assignment will be graded on the execution times and speedups obtained for the two parallel algorithms with respect to a sequential algorithm.